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ABSTRACT
A dual-response surface optimization approach assumes that response surface models of the mean and standard deviation of a response are fitted well to experimental data. However, it is often difficult to satisfy this assumption when dealing with a large volume of operational data from a manufacturing line. The proposed method attempts to optimize the mean and standard deviation of the response without building response surface models. Instead, it searches for an optimal setting of input variables directly from operational data by using a patient rule induction method. The proposed approach is illustrated with a step-by-step procedure for an example case.

KEYWORDS
data mining; dual-response surface optimization; patient rule induction method; process optimization; response surface methodology

Introduction

The response surface methodology (RSM) consists of a group of techniques used to empirically study a response to a number of input variables. Here, a researcher attempts to find the optimal setting for the input variables that either maximizes or minimizes the response (Myers et al. 2009). RSM assumes that the variance of the response is constant and focuses on optimizing the mean response. However, assuming a constant variance might not be valid in practice. In such a case, both the mean response and the standard deviation of the response should be considered when determining the optimum conditions for the input variables.

Dual-response surface optimization (DRSO) attempts to optimize both the mean and standard deviation of the response. A conventional approach to DRSO is to build statistical models for the mean and standard deviation of the response by fitting statistical models to experimental data. Then, an optimal setting for the input variables is obtained by analyzing the statistical models (Lee and Kim 2012).

However, this conventional approach can have two practical limitations. First, efforts to conduct such experiments can be burdensome for process engineers. In order to build the statistical models, experiments must be conducted, and experimental data should be obtained. A large number of experiments are typically conducted for DRSO because replicate experiments at each design point are needed to fit the statistical model for the standard deviation of the response; alternatively, replicate experiments are not mandatory in conventional RSM. Thus, efforts to conduct a large number of experiments can be a burden for process engineers.

Second, the mean and standard deviation of the response at the optimal setting for the input variable have uncertainty (Lee and Lee 2016). This is because the optimal setting is obtained from prediction models. Care must be taken regarding this uncertainty issue if the prediction capability of the prediction models is poor. Because the prediction model varies from the true model, the resulting optimal setting can be far from optimal (Xu and Albin 2003). Also, it has been reported that the prediction capability of prediction models is often poor when the models are fitted to operational data from manufacturing lines whose volume is large (Lee and Kim 2008). In many cases, it is common to have operational data rather than experimental data due to the development of information technologies such as IoT (internet of things) technology and big data analytics in manufacturing lines. Data from equipment in manufacturing lines are automatically gathered and stored in the manufacturing database system; thus, it

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becomes increasingly important to analyze these data for process optimization.

When dealing with the operational data from manufacturing lines, a data mining approach is a good alternative to overcome the aforementioned difficulties. Data mining approaches do not use prediction models; thus, they avoid uncertainty issues. Also, they utilize operational data, which means that no experiments are required. There is a greater need to use data mining in DRSO because RSM-based DRSO requires a large number of experiments, which results in a large experimental cost. Several replicates should be conducted at every design point in order to estimate the variability of the response. Unfortunately, there are no techniques that can be used to replace the process of replicating experiments.

An attractive data mining approach that can be used for process optimization is the patient rule induction method (PRIM) (Friedman and Fisher 1999). PRIM searches a set of sub regions of the input variable space, within which the performance of the response is considerably better than that of the entire input domain (Chong et al. 2007).

PRIM has been adopted for process optimization in the literature. Chong et al. (2007) adopted PRIM for process optimization of the mean of a single response by assuming that the variance of the response was stable. However, the stable variable assumption might be invalid in practice. Lee and Kim (2008) adopted PRIM to optimize multiple responses. This work also neglected the variability of the responses. Overall, no works have adopted PRIM to solve a dual-response problem, where the mean and standard deviation of the response are optimized simultaneously.

The purpose of the proposed method is to optimize the mean and standard deviation of the response variable simultaneously by adopting PRIM. The proposed method is a model-free approach. Thus, it is free from the uncertainty issues that result from model building. Also, it utilizes operation data, which means that it is not necessary to obtain experimental data.

The remainder of the article is organized as follows. Existing DRSO methods and PRIM are reviewed. The proposed method is presented and the proposed method is illustrated with an example. Finally, concluding remarks are made.

Literature review

This section reviews the dual-response surface optimization and PRIM techniques that are most relevant to the proposed method.

Dual-response surface optimization

DRSO attempts to find a set of input variables that simultaneously optimize the mean and standard deviation of the response. In DRSO, the mean and standard deviation of the response are modeled based on experimental data. They are often modeled as quadratic functions of the input variables x_j $(j = 1, 2, \ldots, p)$. The two models for the mean and standard deviation of the response can be represented as follows:

$$\hat{\omega}_\mu(\mathbf{x}) = \hat{\beta}_0 + \sum_{j=1}^{p} \hat{\beta}_j x_j + \sum_{j=1}^{p} \hat{\beta}_{jj} x_j^2 + \Sigma \sum_{j<k}^{p} \hat{\beta}_{jk} x_j x_k\,,$$

$$\hat{\omega}_\sigma(\mathbf{x}) = \hat{\gamma}_0 + \sum_{j=1}^{p} \hat{\gamma}_j x_j + \sum_{j=1}^{p} \gamma_{jj} x_j^2 + \Sigma \sum_{j<k}^{p} \hat{\gamma}_{jk} x_j x_k\,,$$

where $\hat{\omega}_\mu(\mathbf{x})$ and $\hat{\omega}_\sigma(\mathbf{x})$ are the estimated mean and standard deviation of the response, respectively.

DRSO was first presented by Myers and Carter (1973) and then subsequently popularized by Vining and Myers (1990). Vining and Myers (1990) minimized the standard deviation of the response while keeping the mean of the response at a specified target. Lin and Tu (1995) later suggested a way to minimize the mean squared error (MSE). The MSE consists of a squared bias component and a variance component:

$$\mathrm{MSE} = (\hat{\omega}_\mu - T)^2 + \hat{\omega}_\sigma^2,$$

where T is the target of the response.

Most existing DRSO methods can be considered as model-based approaches. The mean and standard deviation of the response are modeled in the form of a response surface based on the RSM framework. However, this model-based approach suffers from the uncertainty issue. Furthermore, it is not appropriate for large volumes of operational data due to its poor prediction capability. Thus, we propose a particular model-free approach for dual-response optimization, called dual-response PRIM (DR-PRIM).

Patient rule induction method

PRIM searches a set of sub-regions of the input variable space within which the performance of the response is considerably better than that of the entire input domain (Chong et al. 2007). Here, the response is considered to be a larger-the-better type response. Although PRIM originally allowed the input variables to have both real and categorical values (Friedman and Fisher 1999; Chong et al. 2007), only input variables with real values are considered here for the sake of simplicity.

For the p input variables, $x_1, x_2, \ldots, x_p$, a p-dimensional box B is defined as the intersection.

$$B = s_1 \times s_2 \times \cdots \times s_p.$$

Here, s_j is a subrange of the jth input variable, x_j, denoted by $s_j = (l_j, u_j)$, where l_j and u_j are the lower and upper limits, respectively.

$$\beta_B = \frac{n_B}{N},$$

Suppose that we have N observations denoted by $\{(y_i, \mathbf{x}_i), \mathrm{i} = 1, 2, \ldots, N\}$, where y_i and $\mathbf{x}_i$ are values of the response and the input variables of the ith observation, respectively. $\mathbf{x}_i$ is a p-dimensional vector denoted by $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$. When $\mathbf{x}_i$ is located in box B (i.e., $l_1 \le x_{i1} \le u_1$, $l_2 \le x_{i2} \le u_2, \ldots, l_p \le x_{ip} \le u_p$), it is denoted by $\mathbf{x}i \in B$.

Given a box B and observations $\{(y_i, \mathbf{x}_i), i = 1, 2, \ldots, N\}$, there are two statistics that describe the properties of box B. The first one is the support β_B, which denotes the proportion of the observations located in box B.where n_B denotes the number of observations that are located inside box B. The support ranges from zero to one. When all of the observations are located in box B, $\beta_B = 1$. The second useful statistic is the box objective Obj_B, which is the mean value of the response in box B.

$$Obj_B = \bar{y}_B = \frac{1}{n_B} \sum_{\mathrm{x}_i \in B} y_i$$

The main idea of PRIM is presented below. Given observations $\{(y_i, \mathbf{x}_i), i = 1, 2, \ldots, N\}$, an initial box B_0 is formed by defining $l_j = \min_{i=1,2,\ldots,N} x_{ij}$ and $u_j = \max_{i=1,2,\ldots,N} x_{ij}$ for $j = 1, 2, \ldots, p$. From B_0, $2p$ candidate boxes, denoted by $\{C_{01\text{-}}, C_{01+}, C_{02\text{-}}, C_{02+}, \ldots, C_{0p\text{-}}, C_{0p+}\}$, are created. The candidate boxes $C_{0j\text{-}}$ and C_{0j+}, $j = 1, 2, \ldots, p$ are obtained by peeling $100\alpha\%$ of the observations in box B_0. The observations are peeled from the edge of box B_0 to the center of box B_0. Figure 1 shows a hypothetical example of $p = 2$ to illustrate the peeling mechanism. The parameter α determines the peeling rate; its value is typically set between 0.05 and 0.1.

Once the $2p$ candidate boxes are obtained, PRIM chooses the one that has the largest box objective among the candidate boxes; this box becomes B_1. If the support of B_1 is greater than a stopping parameter, β_0, then PRIM continues to carry out more peelings to generate the next box B_2. Otherwise, the algorithm ends. This iterative process continues until the support of the box is less than the predetermined value, β_0. The value of β_0 determines the stability of the box. When β_0 is too small, the box often becomes over fitted. According to Friedman and Fisher (1999), it is recommended to set the value of β_0 between 0.05 and 0.1.

Proposed method

In this section, the proposed method, which we refer to as dual-response PRIM (DR-PRIM), is presented. DR-PRIM is a modified PRIM that attempts to optimize the mean and standard deviation of the response. As mentioned in the introduction section, existing model-based approaches for dual-response optimization are limited in that the prediction models have difficulties explaining a large amount of operational data. The prediction capability of the models is often too low; thus, the obtained setting of the input variables can be far from optimal. In contrast, the proposed DR-PRIM is a model-free approach, which avoids the low prediction capability issue.

Figure 2 shows the eight-step procedure of DR-PRIM. Steps 3–7 iterate until the stopping criterion is satisfied. Details of each step are described below.

Step 1. Split all of the data into training and test data

The entire dataset is randomly split into training and test sets. Typically, the size of the training set is two times larger than the size of the test set.

Step 2. Set the values of the parameters

The values of the three parameters are set. The iteration counter m is set as $m = 0$, and the values of α and β_0 are determined. The parameter α determines the peeling rate. Typically, its value is set between 0.05 and 0.1, which is faithful to the original concept of “patient” rule induction. Typically,

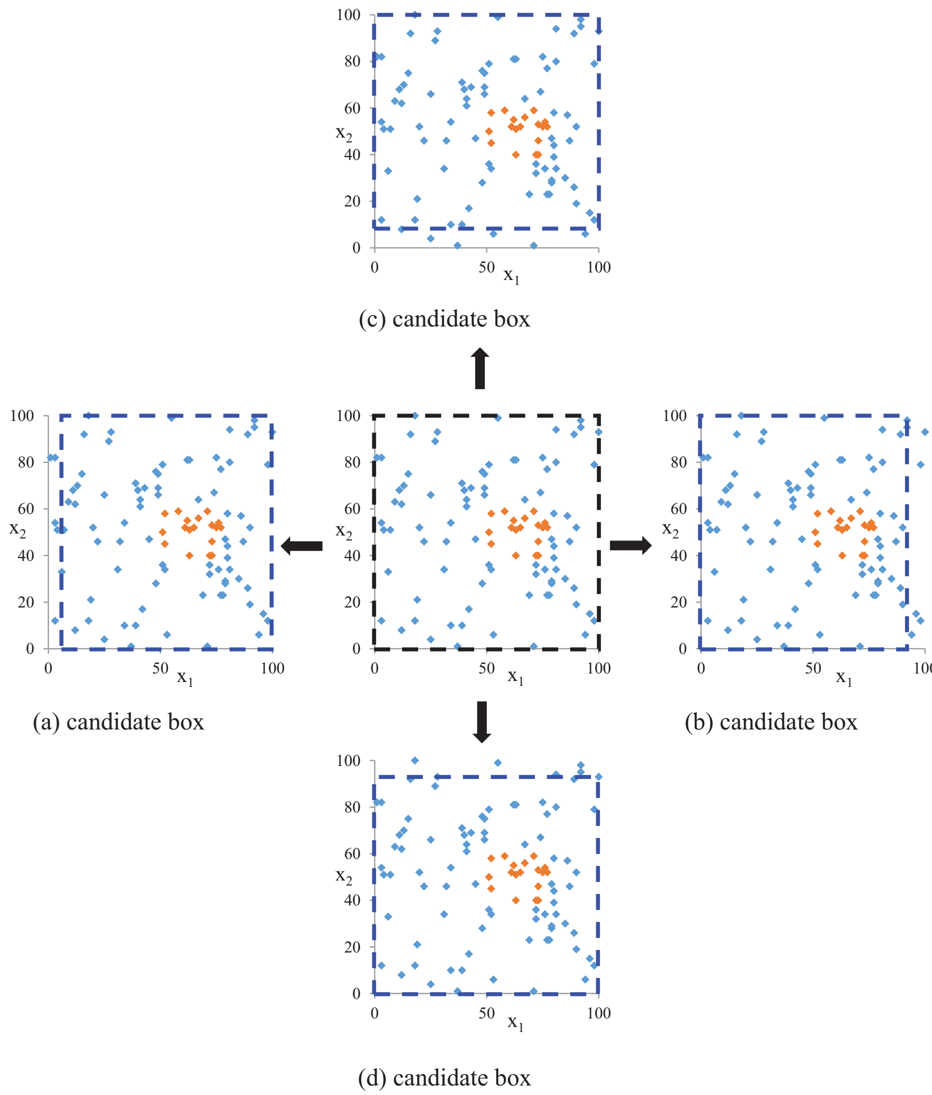

Figure 1. Illustration of the peeling mechanism.

β_0 is set to 0.05. See Friedman and Fisher (1999) for a detailed discussion of the α and β_0 values.

Step 3. Define the box B_m

For the p input variables, $x_1, x_2, \ldots, x_p$, a p-dimensional box, B_m, is defined by

$$B_m = s_{m1} \times s_{m2} \times \cdots \times s_{mp},$$

where $s_{mj} = [l_{mj}, u_{mj}]$ is a subrange of xj, $j = 1, 2, \ldots, p$. When $m = 0$, the box Bm contains all observations of the training data $\{(yi, xi)$, i = 1, 2, $\ldots, N\}$. Thus, the subranges are defined by $l_{0j} = \min_{i=1,2,\ldots,N} x_{ij}$, $u_{0j} = \max_{i=1,2,\ldots,N} x_{ij}$, for j = 1,2, $\ldots, p$. As the iteration counter m increases, these subranges are updated.

Step 4. Generate the candidate boxes

For x_j, two candidate boxes (denoted by C_{j-} and C_{j+}) are generated by peeling $100\alpha\%$ of the observations in box $B_{m.}$ As a result, a total of $2p$ candidate boxes are generated, as shown below.

$$C_{j-} = s_{m1} \times \cdots \times s_{mj}^{(-)} \times \cdots \times s_{mp},$$
$$C_{j+} = s_{m1} \times \cdots \times s_{mj}^{(+)} \times \cdots \times s_{mp},\ j = 1, 2, \cdots, p,$$

where $s_{mj}^{(-)} = [x_{j(\alpha)}, u_{mj}]$ and $s_{mj}^{(+)} = [l_{mj}, x_{j(1-\alpha)}]$. $x_{j(\alpha)}$ is the α-percentile of xj in box Bm (i.e., $\Pr(x_j < x_{j(\alpha)} | \mathrm{x} \in B_m) = \alpha$.

Step 5. Calculate the box objective

The box objective of each candidate box is calculated. We suggest using the mean squared error (MSE) as the box objective. MSE is a sum of the bias and variance of the data within the box. By using MSE as the box objective, the variance is also considered in the optimization. This is one of the main differences between our proposed method and the original PRIM. MSE has been employed as an optimization criterion in various DRSO researches,

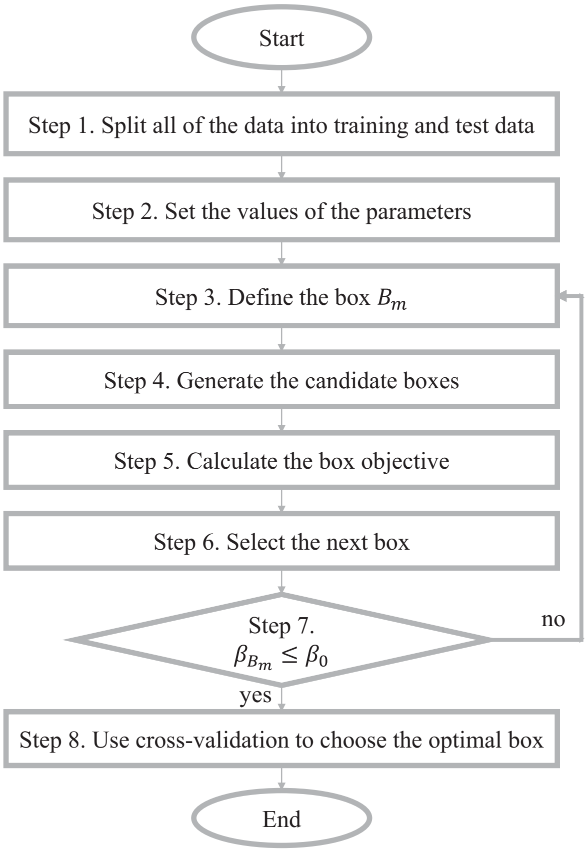

Figure 2. Overall procedure of DR-PRIM.

including studies by Lin and Tu (1995), Ding et al. (2004), Jeong and Kim (2005), and Lee and Kim (2012).

The box objectives of C_{j-} and C_{j+} (for $j = 1, 2, \ldots, p$) are calculated by

$$Obj_{C_{j-}} = (\bar{y}_{C_{j-}} - T)^2 + {s_{C_{j-}}}^2,$$

$$Obj_{C_{j+}} = (\bar{y}_{C_{j+}} - T)^2 + {s_{C_{j+}}}^2.$$

Here, $\bar{Y}_{C_{j-}}$ and $\bar{Y}_{C_{j+}}$ are the sample means of the responses in boxes C_{j-} and C_{j+}, respectively. And T is the target value of the response. Similarly, ${s_{C_{j-}}}^2$ and ${s_{C_{j+}}}^2$ are the sample variances of the responses in boxes C_{j-} and C_{j+}, respectively.

Step 6. Select the next box

In Step 5, a total of $2p$ box objectives (i.e., MSE) are obtained. It should be noted that a smaller value of MSE is preferred. Therefore, the candidate box with the smallest box objective is selected as the next box. Once the next box is determined, m is increased by 1 (i.e., $m = m + 1$), and the next box is denoted as B_m.

Step 7. Check to see if the stopping criterion is satisfied

If the support of box B_m, β_{B_m}, is smaller than the predetermined threshold, β_0, the algorithm ends with box B_m. Otherwise, the algorithm returns to Step 3 to conduct another peeling process. Then, the next box is generated. This iterative procedure continues until the stopping criterion is satisfied. M represents the value of the iteration counter when the stopping criterion is satisfied.

Step 8. Use cross-validation to choose the optimal box

In order to avoid over-fitting problems, cross-validation is conducted. For each B_m ($m = 0, 1, \ldots, M$), the box objective is recalculated with the test data. The optimal box is selected from these boxes by considering both the box objective values from the training data and the recalculated box objective values from the test data. Usually, the recalculated box objective values from the test data are worse than the box objective values from the training data; this is the case because the boxes are obtained from the training dataset only. When the recalculated box objective value is much worse, it is concluded that the boxes are over-fitted to the training dataset. In such a case, it is necessary to check whether outlier data exist in the training and test datasets; outlier data can distort the box objective value (Hastie et al. 2009). Alternatively, new boxes need to be obtained by changing the values of the parameters (α and β_0).

Example: Concrete compressive strength problem

Problem description

In this section, the proposed DR-PRIM is illustrated via an example scenario, i.e., the concrete compressive strength problem (Yeh 1998). The response is concrete compressive strength (y), which is affected by eight input variables ($x_1, x_2, \ldots, x_8$) including the cement, blast furnace slag, and so on. The purpose of this case study is to keep the mean of y at its target value of 60 (i.e., $T = 60$), while also minimizing the standard deviation of y by controlling the input variables.

Optimization of the concrete compressive strength problem

In this section, the optimization is explained by a step-by-step procedure.

Table 1. Subranges of input variables.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
l_{0j}	102	0	0	121.8	0	801.0	594.0	1.0
u_{0j}	540	359.4	200.1	246.9	32.2	1134.3	992.6	365.0

Step 1. Split all of the data into training and test sets

The total number of observations is 1030. We split the entire data into training and test datasets. The number of observations for each set is 700 and 330, respectively.

Step 2. Set the values of the parameters

The iteration counter, m, is set as $m = 0$. Additionally, we set $\alpha = 0.05$ *and* $\beta_0 = 0.05$, which are typical values.

Step 3. Define the box B_m

Because $m = 0$, the initial box B_0 from the training data is set as the interaction of the subranges of the eight input variables (i.e., $B_0 = s_{01} \times s_{02} \times \cdots \times s_{08}$). The subrange of each input variable is given in Table 1.

Step 4. Generate the candidate boxes

For each input variable, two candidate boxes are generated. A total of 16 candidate boxes, i.e., $C_{1-}, C_{1+}, C_{2-}, C_{2+}, \ldots, C_{8-}, C_{8+}$, are generated from B0. C_{3-} and C_{3+} are given below as examples.

$$C_{3-} = s_{01} \times \cdots \times s_{03}^{(-)} \times \cdots \times s_{08},$$
$$C_{3+} = s_{01} \times \cdots \times s_{03}^{(+)} \times \cdots \times s_{08},$$

where $s_{03}^{(-)} = [x_{3(0.05)}, u_{03}]$ and $s_{03}^{(+)} = [l_{03}, x_{3(0.95)}]$.

Step 5. Calculate the box objective

The box objectives of the 16 candidate boxes are calculated. These are $Obj_{C1-} = 832.07$, $Obj_{C2-} = 862.99, \ldots, Obj_{C8-} = 714.45$, $Obj_{C1+} = 899.77$, $Obj_{C2+} = 846.41, \ldots,$ and $Obj_{C8+} = 893.63$.

Step 6. Select the next box

The box with the smallest box objective is C_{8-}. Thus, C_{8-} is selected as the next box. The iteration counter m is increased by 1 (i.e., $m = 1$), and the selected box is defined as B_1. Its box objective is smaller than Obj_{B0} (i.e., 714.45< 866.62). It should be noted that the support of B_1 is $\beta_{B_1} = 0.63$. Usually, β_{B_1} is 0.95 when the peeling rate is 0.05. However, in this particular example, 37% of the observations were peeled off in the first iteration; thus, β_{B_1} becomes 0.63. This occurred because many observations have the same value of x_8; these were all excluded by a single peeling process.

Table 2. Results of training and test sets for iterations 1 to 21.

	Training set			Test set		
Iteration	MSE	Mean	Variance	MSE	Mean	Variance
1	714.45	39.28	285.20	670.68	39.92	267.47
2	522.85	44.30	276.42	588.62	42.40	278.94
3	419.95	46.93	249.13	498.05	44.82	267.54
4	393.36	47.76	243.54	482.78	45.35	268.12
5	366.63	48.66	238.15	425.59	47.12	259.78
6	327.52	49.77	222.96	385.28	48.45	251.89
7	285.88	50.96	204.24	317.45	50.20	221.36
8	267.28	51.78	199.73	289.60	51.29	213.78
9	250.19	52.42	192.79	265.94	52.79	213.99
10	235.88	53.10	188.32	250.11	53.22	204.19
11	219.93	53.78	181.21	228.68	54.33	196.49
12	205.85	55.51	185.71	186.46	56.58	174.80
13	188.88	56.16	174.13	173.00	57.09	164.52
14	124.37	65.56	93.44	98.41	65.64	66.60
15	110.03	64.88	86.26	110.85	65.63	79.13
16	103.17	64.53	82.68	110.85	65.63	79.13
17	101.12	64.95	76.62	113.40	65.42	84.02
18	**97.25**	**65.14**	**70.85**	138.77	66.48	96.74
19	98.64	65.10	72.65	150.06	66.48	96.74
20	100.52	65.79	67.04	150.06	66.48	96.74
21	98.24	66.77	52.39	150.06	66.48	96.74

Step 7. Check to see if the stopping criterion is satisfied

Because $\beta_{B_1} > \beta_0$ (i.e., 0.63 > 0.05), the algorithm goes back to Step 3. Then, peeling and candidate box selection are conducted to generate the next box B_2. This iterative process continues until the support is less than 0.05. At Iteration 21, the support is less than 0.05; thus, the procedure is ended. As a result, a total of 21 boxes (i.e., $B_1, B_2, \ldots, B_{21}$) were obtained.

Step 8. Use cross-validation to choose the optimal box

We conducted cross-validation on the 21 optimal boxes. The box objective values of the 21 boxes were recalculated from the test data. Table 2 summarizes the box objective values obtained from both the training data (left side) and test data (right side). We selected B_{18} as the optimal box because it had the smallest box objective value among the 21 boxes (i.e., $Obj_{B_{18}}$= 97.25). Although the recalculated box objective value of B_{18} was larger than 97.25, we concluded that the gap between the two box objective values was not significant.

Effect of the random split

In Step 1, all of the data are randomly split into the training and test datasets. Thus, the result of the optimization depends on the random split, and it is desirable that optimization be robust to the effect of the random split. In order to investigate the effect of the random split, we solved the problem several times by randomly splitting all of the data into the training

Table 3. Optimization results for the random split by DR-PRIM.

Replication	Training set $\bar{y}$	Training set s_y^2	Test set $\bar{y}$	Test set s_y^2
1	65.14	70.85	66.48	96.74
2	62.02	89.96	65.61	81.31
3	66.42	58.72	66.10	98.49
4	65.16	72.57	67.81	70.86
5	65.83	65.54	68.42	72.40

Table 4. Optimization results of the example case with different values of α and β_0.

Case	α	β_0	$\bar{y}$	s_y^2
1	0.05	0.01	58.53	12.15
2	0.05	0.05	65.14	70.85
3	0.05	0.1	61.22	119.40
4	0.1	0.01	62.00	3.67
5	0.1	0.05	61.33	73.02
6	0.1	0.1	60.64	127.30

and test datasets. We conducted five replications and compared the results. In each replication, as was done in the previous section, we obtained the optimal box from the training data and then recalculated the box objective value from the test data.

Table 3 compares the box objective values of the optimal boxes for the five replications. The comparison results indicate that the five mean and variance values of the responses are not significantly different. Therefore, DR-PRIM is believed to obtain a stable solution for the random split in this example.

Effects of α and β_0

The determination of the initial parameters α and β_0 is important because the optimization results depend on these two parameters. In order to investigate the effects of α and β_0, we conducted a sensitivity analysis with respect to these two parameters. We considered six combinations of α and β_0. The results of the sensitivity analysis are shown in Table 4.

It should be noted that the effect β_0 is quite large compared to that of α. The variance of the response decreases remarkably as β_0 decreases, while the mean of the response shows a small change in value. Consequently, the box objective (i.e., MSE) decreases as β_0 decreases. This phenomenon indicates that a tradeoff exists between the performance and stability of the proposed method. When a small value of β_0 is applied, the iteration of DR-PRIM continues until a very small number of observations remain. This box would have a small MSE value; however, the box would be unstable and would not be robust to the effects of process variations because the size of the box is small. Thus, it is important to choose an appropriate parameter according to the analytic goal (Friedman and Fisher 1999).

Comparison of DR-PRIM with the original PRIM

For further analysis, the proposed DR-PRIM is compared with the original PRIM proposed by Friedman and Fisher (1999). The original PRIM only considers the mean of the response variable. Table 5 compares the optimization results of PRIM and DR-PRIM. The mean of the response obtained by PRIM is 60.02, which is very close to the target value of 60; however, the variance is 155.73, which is much larger than that of DR-PRIM. As a result, the confidence interval is wider than that of DR-PRIM. This means that the uncertainty of the response value obtained by PRIM is larger than that of DR-PRIM. Similarly, the MSE obtained by PRIM is 155.73, which is larger than that of DR- PRIM (i.e., $155.73 > 97.25$).

We plotted the peeling trajectories of the two methods to determine the changes in the mean and variance with respect to the support. Figures 3 and 4 show the peeling trajectories of PRIM and DR-PRIM, respectively. In the case of the mean (Figure 3), the mean value of PRIM approaches the target value of 60 rapidly as the support changes from 0.63 to 0.1. Then, it slowly approaches the target value as the support decreases from 0.1; it stops at 60.14 when the support reaches 0.05. In contrast, the mean value of DR-PRIM continuously increases as the support decreases.

As shown in Figure 4, the variance value of PRIM decreases as the support decreases from 0.63 to 0.1. However, it increases rapidly as the support decreases from 0.1. This phenomenon implies that the boxes are unstable when the support is less than 0.1. In contrast, the variance value of DR-PRIM continuously decreases

Table 5. Comparison of DR-PRIM and PRIM.

Method	x^*	$\bar{y}$	s_y^2	95% confidence interval
PRIM	$165 \le x_1 \le 446, 13.6 \le x_2 \le 189, x_3 \le 172.4, 4.1 \le x_5 \le 16.5, 852.1 \le x_6 \le 1081, 707.9 \le x_7 \le 887.1, 56 \le x_8 \le 100$	60.02	155.73	[52.19, 67.85]
DR-PRIM	$213.7 \le x_1 \le 491, 24 \le x_2 \le 262.2, 3.9 \le x_5 \le 18.6, 56 \le x_8 \le 100$	65.14	70.85	[61.67, 68.61]

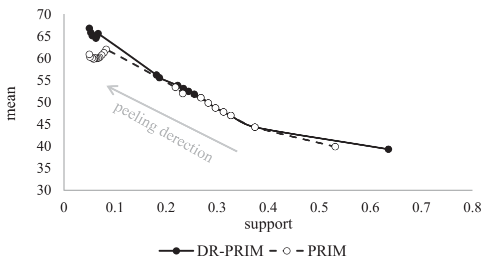

Figure 3. Peeling trajectory for the example case by DR-PRIM and PRIM (mean vs. support).

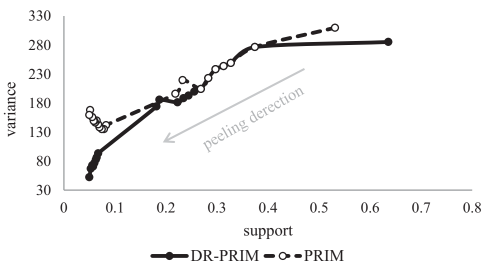

Figure 4. Peeling trajectory for the example case by DR-PRIM and PRIM (variance vs. support).

as the support decreases, which means that the boxes obtained by DR-PRIM are more stable.

Comparison of DR-PRIM with the regression modeling approach

The proposed DR-PRIM is compared with the regression modeling approach. The regression modeling approach is a typical approach when using RSM. In this approach, a statistical model is fitted to the observations, and the optimal settings of the input variables are obtained by analyzing the fitted model. We fitted a second-order model to the entire data set (i.e., 1030 observations). The fitted model is given below.

$$\begin{aligned}\hat{y} = & -4613 + 2.61\,x_1 + 2.20\,x_2 + 1.67\,x_3 \\ & + 14.4\,\mathrm{x}_4 + 31.0\,x_5 + 2.69\,x_6 + 3.72\,x_7 \\ & + 0.080\,x_8 - 0.000354\,x_1^2 - 0.000268\,x_2^2 \\ & + 0.000164\,x_3^2 - 0.00902\,x_4^2 - 0.0453\,x_5^2 \\ & - 0.000361\,x_6^2 - 0.000833\,x_7^2 - 0.000615\,x_8^2 \\ & - 0.000508\,x_1x_2 - 0.000194\,x_1x_3 \\ & - 0.00467\,x_1x_4 - 0.0109\,x_1x_5 - 0.000634\,x_1x_6 \\ & - 0.000936\,x_1x_7 + 0.000199\,x_1x_8 \\ & + 0.000101\,x_2x_3 - 0.00397\,x_2x_4 - 0.0104\,x_2x_5 \\ & - 0.000616\,x_2x_6 - 0.000714\,x_2x_7 \\ & + 0.000436\,x_2x_8 - 0.00449\,x_3x_4 - 0.0174\,x_3x_5 \\ & - 0.000332\,x_3x_6 - 0.000470\,x_3x_7 \\ & + 0.000700\,x_3x_8 - 0.0337\,x_4x_5 - 0.00498\,x_4x_6 \\ & - 0.00571\,x_4x_7 - 0.000080\,x_4x_8 \\ & - 0.0101\,x_5x_6 - 0.0122\,x_5x_7 + 0.00337\,x_5x_8 \\ & - 0.00101\,x_6x_7 - 0.000056\,x_6x_8 \\ & + 0.000246\,x_7x_8.\end{aligned}$$

The R^2 and adjusted R^2 of the fitted model are 81.1% and 80.2%, respectively. Next, we obtained the optimal settings for the input variables by solving the following optimization problem.

$$\begin{aligned}&\underset{\mathbf{x}}{\text{minimize}}\ |\hat{y} - 60|, \\ &\text{subject to}\ \ l_{0j} \le x_j \le u_{0j}, \quad \text{for } j = 1, 2, \ldots, 8.\end{aligned}$$

The obtained optimal setting is $\mathbf{x}* = (115.04, 9.78, 0.24, 198.25, 32.20, 812.91, 611.27, 8.05)$. At this setting, the expected response value is 60 (i.e., $\hat{y} = 60$), and the 95% confidence interval is $[-53.13, 172.99]$. Although the regression modeling approach achieved the target value of 60, the estimated value at the optimal condition is quite unstable because the range of the confidence interval is too wide. In contrast, the confidence interval at B_{18}, obtained by the proposed DR-PRIM, is sufficiently narrow.

The above regression approach neglects the variability of the response, thus, a model for the variance of the response is built for a fair comparison. The regression model for the variance can be built when replicated observations exist. We found that only 34 observations were replicated and these observations were located at 8 design points. We concluded that the variance modeling approach is not good for this case due to the insufficient number of replicated observations.

Alternatively, we excluded x_8 in order to obtain more replicated observations. We selected x_8 for the exclusion because it has the smallest effect among the eight input variables according to the regression analysis. As a result, we found that 602 data were replicated and these data were located at 182 design points. At each design point, we calculated the sample variance and built a linear regression model by using the 182 sample variances. The R^2 of the fitted model is 38.71% and it shows high variance inflation factor values. This statistical information implies that the prediction capability is low, thus, the variance modeling is not appropriate for this case study. As we observed in the case study, applying the regression modeling approach to the operational data is not a good strategy due to the low prediction capability. It has been reported that the prediction capability of prediction models is often poor when the models are fitted to operational data whose volume is large (Lee and Kim, 2008).

Trade-off between mean and variance in DR-PRIM

The mean and variance of the response variable are often in conflict. In this section, we run the DR-PRIM with different MSE weights in order to investigate the relationship between the mean and variance. We use the weighted MSE (WMSE) as a new box objective. According to Ding et al. (2004), using WMSE as an optimization criterion is a good way to balance the mean and variance. The new box objective is as follows:

$$Obj_{B_m} = \lambda(\bar{y}_{B_m} - T)^2 + (1-\lambda){s_{B_m}}^2.$$

We increased the weight factor λ from 0 to 1 in increments of 0.1. Figure 5 shows the mean and variance values of the response according to the λ value. When $\lambda = 1$, only the mean of the response is considered in the optimization. Thus, the results $\bar{y} = 60.02$ and $s_y^2 = 155.73$ are the same as those obtained by PRIM, as shown in Table 5. When $\lambda = 0$, only the variance of the response is considered; thus, a small variance of 53.41 was achieved. However, the mean of the response is 29.42, which is far from the target value of 60. It should be noted that the gray dot represents an attractive solution; here, the variance is much smaller than that of the solution obtained by applying $\lambda = 0$ and the mean of the response is near 60. This implies that the variance can be considerably improved if a small amount of bias is allowed. This WMSE-based approach would help process engineers investigate the tradeoffs between the mean and variance of the response.

On the other hand, process capability indices such as Cp and Cpk can be used for comparing several solutions of different weight values. Here, we assume that

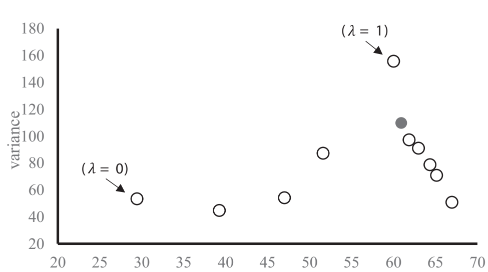

Figure 5. Mean and variance with different values of the weight factor λ.

Table 6. Process capability index for several solutions.

sλ	$\bar{y}$	s_y^2	Cp	Cpk
0	29.42	53.41	1.37	− 0.03
0.1	39.24	44.80	1.49	0.46
0.2	47.00	54.19	1.36	0.77
0.3	51.60	87.31	1.07	0.77
0.4	66.97	50.83	1.40	1.08
0.5	65.14	70.85	1.19	0.98
0.6	64.36	78.77	1.13	0.96
0.7	62.98	91.02	1.05	0.94
0.8	61.87	97.24	1.01	0.95
0.9	60.93	109.80	0.95	0.92
1	60.02	155.70	0.80	0.80

lower and upper specification limits for y are 30 and 90, respectively. Table 5 reports Cp and Cpk values for the solutions in Table 6. From the perspective of process capability index, the solution obtained from $\lambda = 0.4$ is an attractive solution because both of Cp and Cpk have large value compared to the other solutions.

Concluding remarks

In this work, we proposed a modified version of PRIM, which we refer to as DR-PRIM. DR-PRIM attempts to optimize the mean and variance without building response surface models. It obtains an optimal setting for the input variables directly from the operational data, in which the mean and variance are optimized. We illustrated DR-PRIM with an example case and compared the optimization results with those of PRIM and regression modeling approaches. It was shown that DR-PRIM works well and is advantageous in finding the optimal box, where the mean of the response is close to the target (i.e., a small bias) and the variance is low.

The key idea of the proposed method is to use the MSE as the box objective in PRIM. Not only MSE but also other functions such as desirability function can be as the box objective in DR-PRIM. The desirability function was first suggested by Derringer and Suich (1980) and it has been applied to many multiresponse optimization problems. Because the dual response problem can be considered as the special case of the multiresponse problem, the desirability function can be also used in DR-PRIM. The desirability function employs various preference parameters, which means the preference of process engineers for the mean and standard deviation can be easily incorporated into the optimization process. Thus, optimization can be conducted in a flexible manner according to the preferences of the process engineers.

When applying the desirability function, two functions d_μ and d_σ should be modeled first. The functions d_μ and d_σ represent the desirability for the mean and standard deviation of the response, respectively. They can be modeled as follows

$$d_\mu(\bar{y}) = \begin{cases} 0 & \text{if } \bar{y} \leq \mathrm{LSL}_\mu \text{ or } \bar{y} \geq \mathrm{USL}_\mu \\ \left(\frac{\bar{y}-\mathrm{LSL}_\mu}{T-\mathrm{LSL}_\mu}\right)^{s_\mu} & \text{if } \mathrm{LSL}_\mu \leq \bar{y} \leq T \\ \left(\frac{\mathrm{USL}_\mu-\bar{y}}{\mathrm{USL}_\mu-T}\right)^{t_\mu} & \text{if } T \leq \bar{y} \leq \mathrm{USL}_\mu \end{cases},$$

$$d_\sigma(s_Y) = \begin{cases} 1 & \text{if } s_Y \leq \mathrm{LSL}_\sigma \\ \left(\frac{\mathrm{USL}_\sigma-s_Y}{\mathrm{USL}_\sigma-\mathrm{LSL}_\sigma}\right)^{t_\sigma} & \text{if } \mathrm{LSL}_\sigma \leq s_Y \leq \mathrm{USL}_\sigma, \\ 0 & \text{if } s_Y \geq \mathrm{USL}_\sigma \end{cases}$$

where LSL_μ and USL_μ are lower and upper specification limits for $\bar{y}$, respectively. Similarly, LSL_σ and USL_σ are lower and upper specification limits for s_Y, respectively. Parameters s_μ, t_μ, and t_σ determine shape of the desirability functions. All of these parameters are determined according to the preferences of the process engineers. Once the two desirability functions are modeled, they are aggregated into a single measure, called overall desirability function. Geometric mean is often used as the aggregation. The box objective is defined as the overall desirability function.

When modeling the desirability functions, the lower and upper specification limits should be carefully determined. When $\bar{y}$ or s_Y fall outside the specification limits, the desirability function value becomes 0. When all of candidate box objectives are 0, the next box is selected randomly. If the random selection process occurs in many iterations, the algorithm would be ended with finding local optimum. Usually this random selection occurs in the first iteration because $\bar{y}$ and s_Y values of entire observations are poor, thus, they often fall outside the specification limits. Therefore, in order to prevent the box objective from having 0, the specification limits need to be set wide enough.

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